

Hassan Jamal Khan

AI/ML Engineer

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Portfolio: hassankhan86.github.io/Portfolio

Summary

AI/ML Engineer with 3+ years of software engineering experience, specializing in Machine Learning, Deep Learning, NLP, Generative AI, and LLM-based applications. Hands-on experience building end-to-end ML pipelines, RAG and agentic systems, computer vision solutions, and production deployments using Python, LangChain, Scikit-learn, TensorFlow, Docker, and AWS. Currently pursuing an MS in Data Science, with a solid foundation in statistics and a focus on solving real-world problems with AI.

Technical Skills

- **Programming:** Python, SQL
- **AI/ML:** Scikit-learn, Pandas, NumPy, Matplotlib, Supervised & Unsupervised Algorithms, Feature Engineering, Data Visualization
- **Statistics:** Probability, Hypothesis Testing, Regression Analysis, Statistical Analysis
- **NLP & DL:** NLTK, ANN, CNN, RNN, LSTM, GRU, Encoder-Decoder, Transformers (Keras / TensorFlow)
- **GenAI:** LangChain, RAG, Agentic AI, Crew AI, LangGraph, Hugging Face, Fine-tuning, Quantization
- **Development:** Django, Django REST Framework, Django ORM, JWT Authentication, Celery, Django Channels, Middleware, React, Flask, REST APIs, Streamlit
- **Database:** SQL, MongoDB (PyMongo), Redis, Vector Databases (FAISS, ChromaDB)
- **Deployment & Version Control:** AWS (EC2, ECR, Bedrock, SageMaker), Azure, Docker, Heroku, Git, GitHub
- **Automation & Scraping:** n8n, Selenium, Playwright, BeautifulSoup

Work Experience

LLM Trainer | Apr 2024 – Dec 2025

Turing · Remote

- Worked on multiple LLM training projects, improving model performance through high-quality data labeling, evaluation, and prompt optimization
- Promoted to Team Lead, managing data-quality reviews, coordinating workflows, and tracking team performance to ensure efficient project execution and high-quality outputs

Software Engineer | Nov 2022 – May 2024

TechLeadz · On-site

- Developed and maintained Django web applications integrated with React front-ends, designing and optimizing SQL database schemas to enhance performance and reliability
- Automated Outlook email data ingestion using Python, transforming datasets with Pandas and NumPy based on business logic

Junior Python Developer | Dec 2021 – Sep 2022

Mobylogix · On-site

- Built and deployed RESTful APIs using Django and PyMongo, automating workflows with Celery and Playwright to improve operational efficiency
- Supported full-cycle development, including design, implementation, testing, and deployment

Education

MS in Data Science | 2025 – 2027

Information Technology University, Lahore, Pakistan

Bachelor of Science in Computer Science | 2017 – 2021

Minhaj University Lahore, Pakistan | CGPA: 3.02/4.0

Projects

Shelf Label Detection & Tracking using YOLO | *Computer Vision* | [GitHub](#)

- Designed and deployed an offline, on-device detection and tracking system using **YOLO11n** to identify and classify retail shelf-edge price labels into **size tiers** (small/medium/large).
- Built an end-to-end YOLO training pipeline in Python (Ultralytics) with CLI scripts for **dataset preparation, model training, evaluation**, and quantized export.
- Designed a **two-layer tracking** pipeline to eliminate double-counting during camera pans, pairing **BoT-SORT** motion compensation with a custom Union-Find velocity-extrapolation resolver for track merging.
- Exported the trained model to **ONNX** without baked-in NMS and integrated it into a Flutter application for real-time, on-device shelf inspection.

Multi-Agent Career AI Assistant | *GenAI* | [GitHub](#)

- Built an end-to-end multi-agent AI career platform with specialized agents for resume analysis, job search, market research, cover letter generation, mock interview system, and career advisory using LLM orchestration
- Designed a **LangGraph**-based **supervisor routing** system and **ReAct**-style tool using **AgentExecutor** for intelligent query delegation, reasoning, and tool invocation
- Integrated web research and job search manual tools for real-time career insights, along with ATS evaluation, skill-gap analysis, cover letter generation, and a mock interview system using structured LLM outputs (**Pydantic**)

Real Estate Price Prediction & Recommendation System | *Regression* | [GitHub](#)

- Built an end-to-end **ML pipeline** for property price prediction, performing data preprocessing, **EDA**, and **feature engineering** (area extraction, categorical encoding, outlier handling, VIF, SHAP, RFE) on real-world housing datasets
- Trained and compared 10+ regression models (XGBoost, Random Forest, Extra Trees, etc.) across multiple encoding strategies (OrdinalEncoder, OneHotEncoder, TargetEncoder, PCA), achieving an **R-squared** of **0.895** with **Optuna-tuned XGBoost**
- Developed a recommendation system using **TF-IDF + cosine similarity** and built interactive **visualizations** (geo maps, scatter plots, etc.) in **Streamlit**
- Deployed application on **AWS EC2** for real-time predictions and recommendations

Bank Loan Default Prediction | *Classification* | [GitHub](#)

- Built an end-to-end Bank Loan Default Prediction system using Machine Learning, including data cleaning, EDA, and advanced feature engineering (**WOE/IV, VIF**)
- Trained and evaluated multiple models (Logistic Regression, Random Forest, XGBoost) with **hyperparameter tuning** (**Optuna**, **RandomizedSearchCV**), selecting the best model using ROC-AUC, KS Statistic, and Gini Coefficient
- Improved performance using **SMOTETomek** for class imbalance, **achieving 94%** recall in detecting loan defaults and minimizing high-risk misclassification

Network Security Prediction System | *Classification* | [GitHub](#)

- Built an end-to-end **MLOps pipeline** (Data Ingestion, Validation, Transformation, Model Training) with MongoDB integration for phishing detection
- Implemented **modular, production-grade architecture** with custom logging, exception handling, and reusable pipeline components using Python and Scikit-learn
- Trained and compared multiple classifier models (Random Forest, Gradient Boosting, Logistic Regression, AdaBoost) with **hyperparameter tuning** (**GridSearchCV**)
- Tracked experiments via **MLflow & DagsHub** and deployed a FastAPI app using **Docker + GitHub Actions** CI/CD on **AWS (ECR, EC2)** for real-time predictions

Certifications

- [Machine Learning with Python](#) - Coursera
- [Supervised Machine Learning: Regression and Classification](#) - DeepLearning.AI
- [Advanced Learning Algorithms](#) - DeepLearning.AI
- [Complete Data Science, Machine Learning, DL, NLP Bootcamp](#) - Udemy
- [Complete Generative AI Course With LangChain and Huggingface](#) - Udemy